

## DATA SCIENCE MSC CURRICULUM

### Program: Data Science Specialization

#### ASSESSMENT TYPES:

E: Exam | PG: Practice Grade | XE: Lecture+Practice with Exam | XPG: Lecture+Practice with Practical Grade

#### === COMPULSORY SUBJECTS (73 Credits) ===

##### --- SEMESTER 1 (26 Credits) ---

IPM-3AATDKDEG: Deep Network Development

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-3AATDSEGE: Introduction to Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-3AATLEAE: Legal and ethical aspects of DS and AI

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-3AATRMGE: Research methodology

1 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 5 Credits

IPM-3AATTAMEG: Topics in Applied Mathematics

2 Lecture, 2 Practice, 0 Laboratory, 1 Consultation | XPG | 5 Credits

##### --- SEMESTER 2 (20 Credits) ---

IPM-3AATDSLAB1: Data Science Lab I

Requirement: Introduction to Data Science

0 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | PG | 4 Credits

IPM-3AATTISE: Introduction to Data Security

2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits

IPM-3AATNMEG: Numerical Methods for Optimization

Requirement: Topics in Applied Mathematics

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-3AATMLEG: Machine Learning

Requirement: Introduction to Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

##### --- SEMESTER 3 (27 Credits) ---

IPM-3AATAAE: Audio Analysis

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits

IPM-3AATDSLAB2: Data Science Lab II

Requirement: Data Science Lab I

0 Lecture, 2 Practice, 0 Laboratory, 4 Consultation | PG | 6 Credits

IPM-3AATALE: Image Analysis

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits

IPM-3AATOSTEG: Open-source Technologies for Data Science

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XE | 6 Credits

IPM-3AATSMGE: Stream Mining

2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits

IPM-3AATIME: Text Mining

2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits

#### === COMPULSORY ELECTIVE SUBJECTS (Select 11 Credits) ===

##### --- SEMESTER 1 ELECTIVES (2 Credits) ---

IPM-2AATPREPG: Preparation course for master studies and developing learning skills

0 Lecture, 3 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits

IPM-2AATSJEG: Statistics for signal processing

2 Lecture, 0 Practice, 1 Laboratory, 1 Consultation | XPG | 4 Credits

IPM-2AATCNAGG: Introduction to Cloud Native Architectures

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XPG | 6 Credits | (Also offered in Sem 3)

IPM-2AATDEEG: Introduction to Data Engineering

2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XPG | 6 Credits | (Also offered in Sem 3)

##### --- SEMESTER 2 ELECTIVES (6 Credits) ---

IPM-2AATTCTDSEG: Commercial Technologies for Data Science  
1 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | XE | 4 Credits  
IPM-2AATDMDBE: Data models and database (Lecture)  
2 Lecture, 0 Practice, 0 Laboratory, 1 Consultation | E | 3 Credits  
IPM-2AATDMDBG: Data models and database (Practice)  
0 Lecture, 0 Practice, 2 Laboratory, 1 Consultation | PG | 3 Credits  
IPM-2AATDAAE: Design and analysis of algorithms  
2 Lecture, 0 Practice, 0 Laboratory, 2 Consultation | E | 4 Credits  
IPM-2AATRSEG: Recommender Systems  
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XPG | 6 Credits  
IPM-2AATTOCEAEG: DevOps and Cloud engineering on Azure  
2 Lecture, 0 Practice, 2 Laboratory, 2 Consultation | XPG | 6 Credits  
IPM-2AATDSG: Introduction to Data Security  
0 Lecture, 2 Practice, 0 Laboratory, 0 Consultation | PG | 2 Credits

**--- SEMESTER 3 ELECTIVES (3 Credits) ---**

IPM-2AATADNDEG: Advanced Deep Network Development  
Requirement: Deep Network Development  
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits  
IPM-2AATNSEG: Network Science  
2 Lecture, 2 Practice, 0 Laboratory, 2 Consultation | XE | 6 Credits

**=== THESIS & ADDITIONAL ===**

Thesis consultation  
30 Credits | Semester: 4

IPM-2AATERASMUS: Erasmus mobility  
Maximum 24 Credits | Semester: 3

Elective Subjects: 6 Credits | Semesters: 1, 2

**=== CREDIT SUMMARY ===**

Compulsory Subjects: 73 Credits  
Compulsory Elective Subjects: 11 Credits  
Elective Subjects: 6 Credits  
Thesis: 30 Credits

**TOTAL DEGREE CREDITS: 120 Credits**

Semester Credit Distribution:

Semester 1: 30 | Semester 2: 30 | Semester 3: 30 | Semester 4: 30